



Blockchain Domain Task

Note:

**You have the choice to attempt either (Task-1 or Task-2).
The weightage for each task is different based on complexity.**

Task-1 (40% Weightage - Fundamental Level)

Develop a Solidity smart contract with the following features:

- A state variable uint256 storedNumber
- A function to update storedNumber
- A function to retrieve storedNumber
- Deploy the contract on Remix using a test network

Additionally, answer the following theoretical questions:

1. What is hashing in blockchain? What is a hash function?
2. What are consensus mechanisms?
3. What is the gas fee?

Task-2 (60% Weightage - Advanced Level)

Build a University Management System using a smart contract that includes functionalities for:

- Student enrollment (stores student details and initializes balance)
- Depositing and withdrawing from student balance
- Course enrollment
- Fee payment
- Grade assignment and upgrades
- Fetching student details
- Viewing grades

Note

- You can use any programming language, but Solidity is recommended for beginners using Remix.
- Extra functionalities are encouraged.
- The deposit function must take the value from the transaction.

Submission Guidelines

- Choose and complete either Task-1 or Task-2 based on your preference.
- Clearly organize your work, including code and required explanations.
- You may include additional functionalities to enhance your submission.

For Task-1:

- Submit a Google Drive link containing:
 - A link to your Solidity smart contract code hosted on GitHub
 - Answers to the theoretical questions

For Task-2:

- Submit a GitHub repository link containing- Solidity smart contract code
- Make sure all shared links are accessible (public/viewable).
- Verify that your code runs without errors before submission.

Wishing You All the Best
ADG VIT